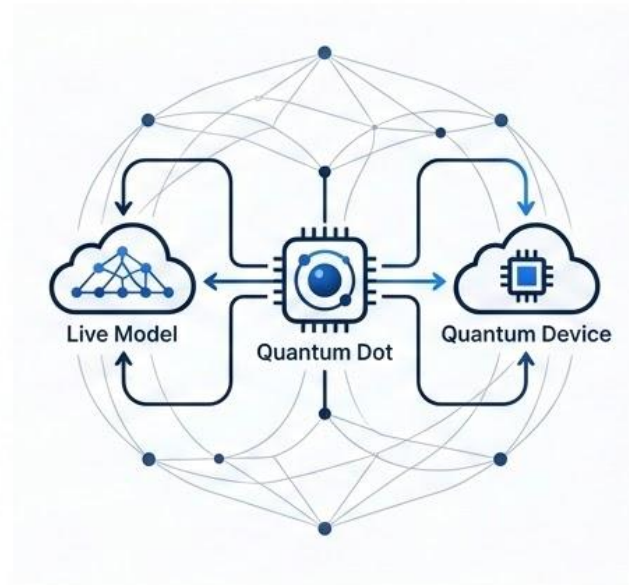


Quantum Dot Live Twin

Keeping a live model synchronised with a quantum device that's changing faster than you can process it.

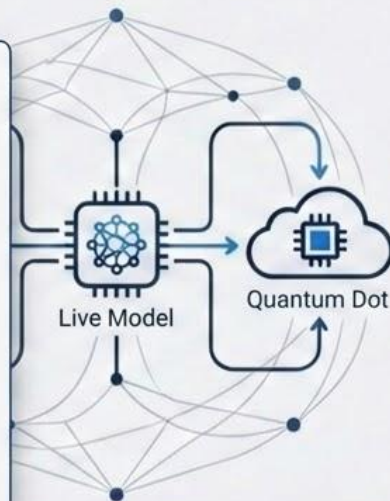


KEY RESULT

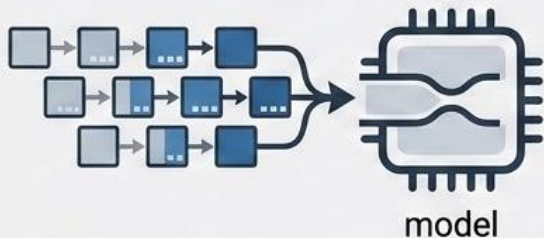
Proven Real-Time Synchronization:

Real qubit readout runs orders of magnitude faster than the models meant to track it. This twin stays synced anyway – measured, not claimed.

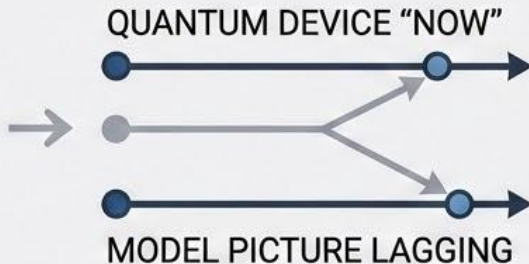
MEASURED, NOT CLAIMED.



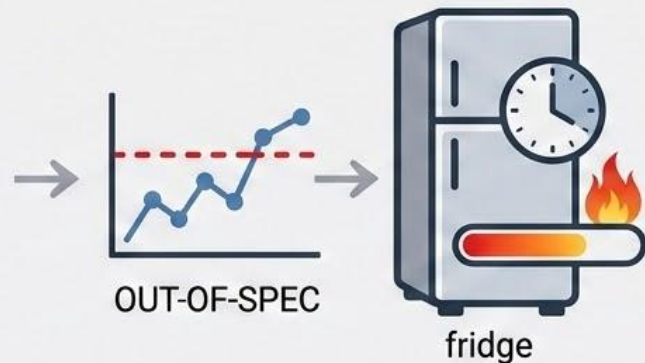
PROBLEM



1. FRAMES ARRIVE
> ABSORPTION RATE



2. SYNCHRONIZATION
LAG GROWS



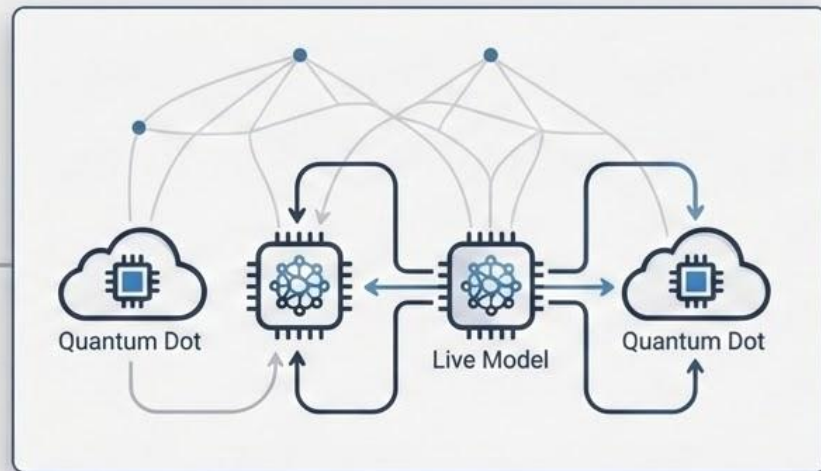
3. DRIFT GOES UNNOTICED;
'FRIDGE TIME' BURNED

SCOPE: WHAT A TWIN IS



1. DOESN'T STEER THE DEVICE

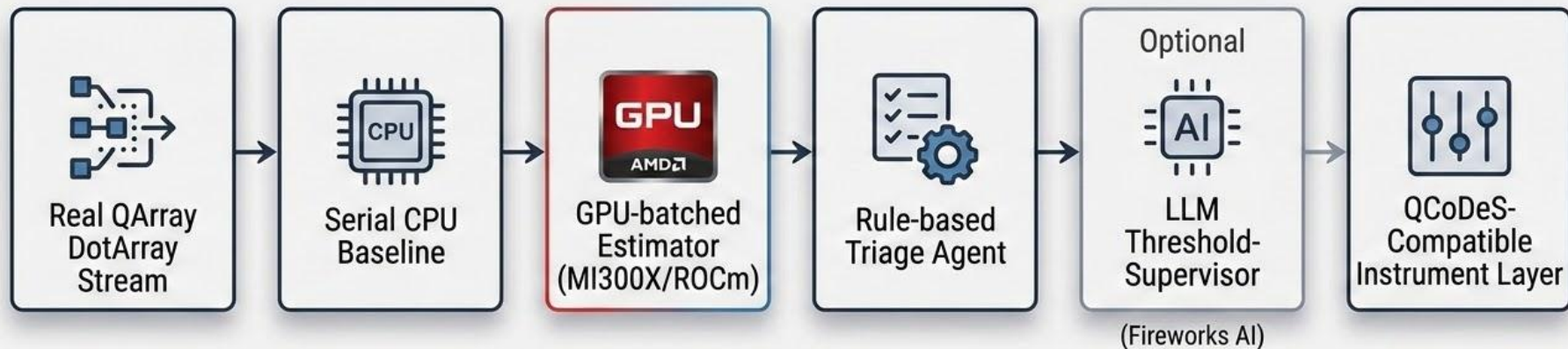
Mirrors its current state.
Stated scope boundary.



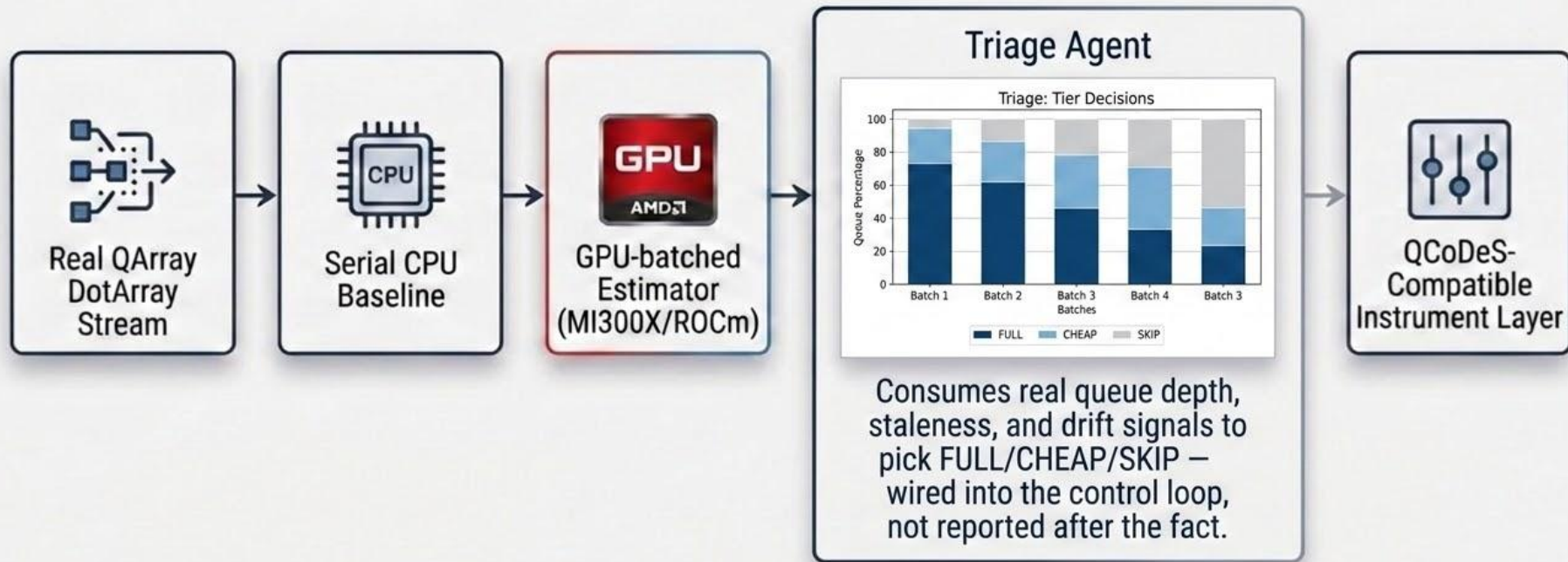
2. SUCCESS IS STAYING SYNCHRONISED

Measured, not claimed.

ARCHITECTURE: REAL-TIME OPTIMIZATION FLOW

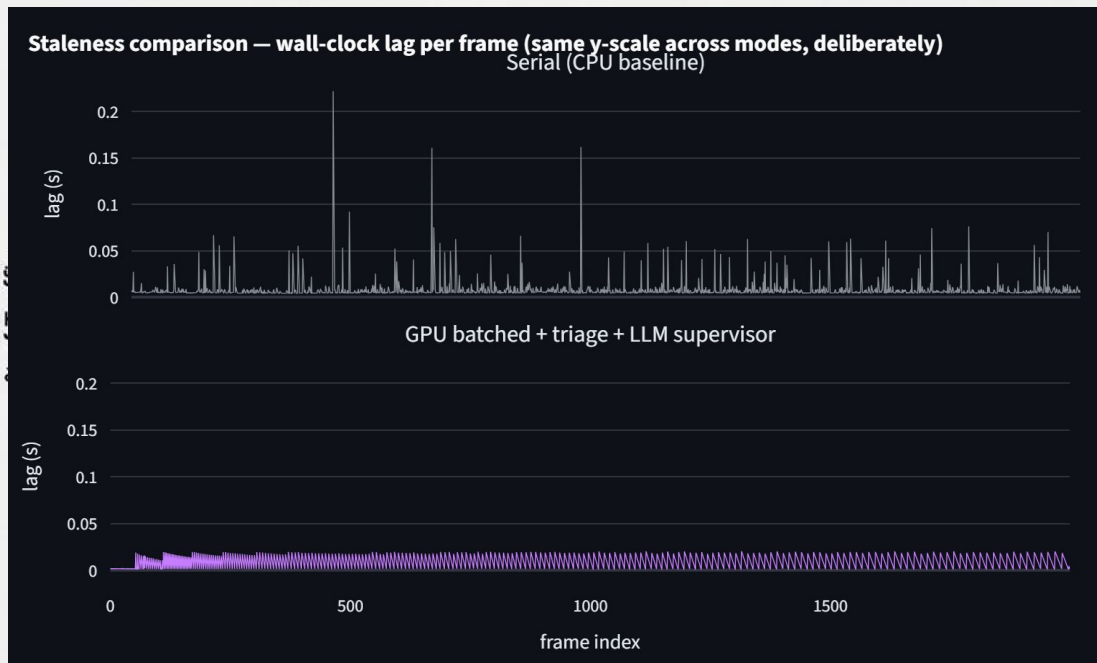


ARCHITECTURE: REAL-TIME OPTIMIZATION FLOW



ARCHITECTURE: THE TRIAGE AGENT REBUTTAL

- CONSUMES REAL SIGNALS (Queue Depth, Staleness, Drift).
- WIRED INTO THE CONTROL LOOP.
(Not reported after the fact.)
- DECISION PATH:
PICK FULL/CHEAP/SKIP.



ARCHITECTURES: BATCHED GPU PERFORMANCE GAIN

PEAK
THROUGHPUT
GAIN



10x - 57x

Faster than
Serial CPU

PLATFORM
USED



AMD MI300X

Batched Inference
(ROCm)

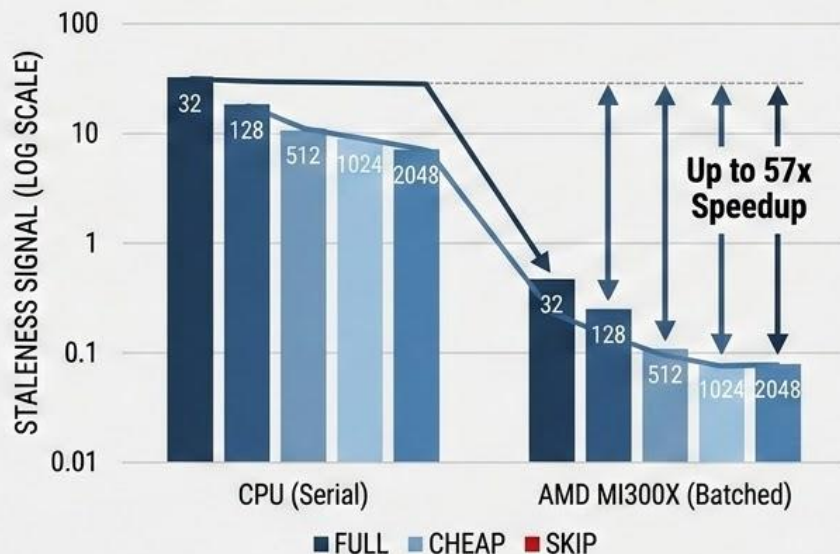
BATCH SIZE
RANGE



32 - 2048

Tested
Workloads

MEASURED STALENESS COMPARISON



PRODUCT/MARKET POTENTIAL

QCoDeS Compatibility

